

Titanic survival prediction

Assignment 1 · From Dirty Data to Predictive Models

An end-to-end comparison of Bernoulli Naive Bayes and linear models using the supplied Kaggle Titanic data.

Colab notebook link: PENDING STUDENT UPLOAD. Upload AppliedML_hw1.ipynb to Colab, enable sharing, and replace this notice with its actual URL before submission.

Main finding

A Ridge model with regularization strength 1 was selected using five-fold cross-validation on the training partition. It correctly classified 143 of 179 held-out passengers (79.9%), compared with 61.5% for a majority-class baseline. Its survivor recall was 69.6% and ROC AUC was 0.832. Ordinary Linear Regression and the selected LASSO model produced the same held-out classifications; the evidence does not establish a practical advantage for Ridge over those two fits.

Bernoulli Naive Bayes reached 70.4% accuracy. Additive smoothing changed its probability estimates slightly but did not change any held-out classifications. Feature engineering improved Linear Regression validation accuracy and reduced Naive Bayes validation accuracy, illustrating that adding plausible predictors does not guarantee improvement.

Page map

Requirement	Report location
Introduction and dataset description	Pages 1–2
Step 1 · Data cleaning	Pages 2–3
Step 2 · Feature engineering	Page 4
Step 3 · Model training	Pages 5–6
Step 4 · Model evaluation	Pages 7–9
Step 5 · Interpretation and limitations	Pages 9–10; synthesis throughout
AI tool usage disclosure and references	Page 11

All numerical findings and figures were generated by the accompanying executed notebook. The report contains analysis rather than code. The holdout was reserved before fitting preprocessing or selecting model settings.

Data source and initial audit

The task is to predict whether a passenger survived, using the historical passenger attributes supplied in Kaggle’s Titanic competition [1]. The local training file contains 891 labeled records and 12 columns, including the binary target Survived. Of these passengers, 342 survived (38.4%) and 549 did not (61.6%). The class imbalance makes accuracy alone incomplete: predicting no survivors already appears moderately accurate.

File	Role in this assignment
train.csv · 891 rows	Labeled development data, split into 712 training and 179 held-out passengers.
test.csv · 418 rows	Unlabeled competition data used only for optional final predictions.
gender_submission.csv	Example prediction format, not observed survival outcomes. Excluded from evaluation.

The predictors describe passenger class, recorded sex, age, accompanying relatives, fare, cabin, embarkation port, and identifiers. Pclass is a category, not a continuous measurement with necessarily equal spacing. SibSp and Parch are counts. PassengerId has no intended predictive meaning. Names and ticket strings are high-cardinality fields that could encourage memorization in this small sample.

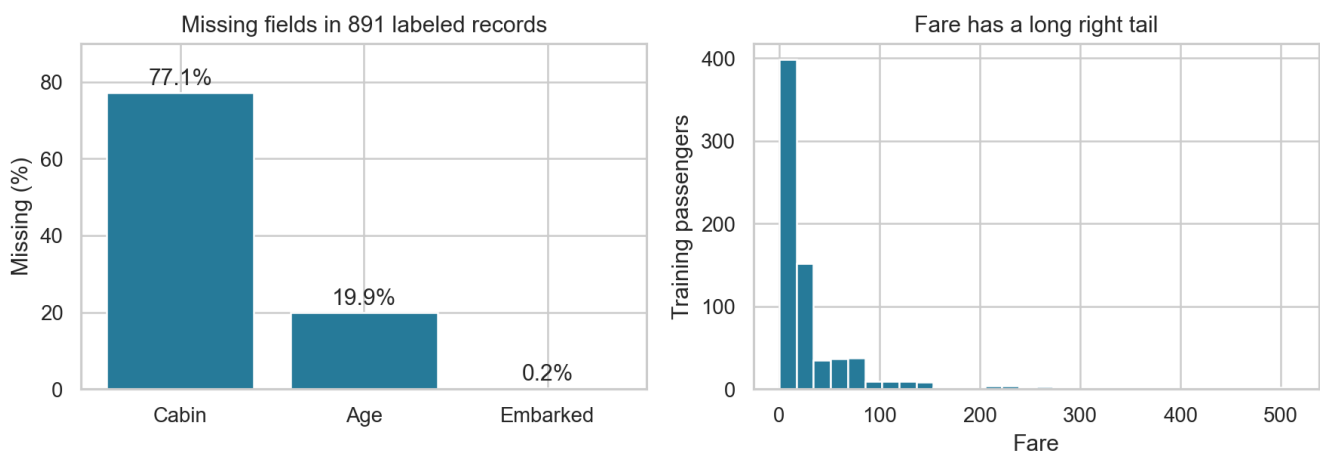


Figure 1. Missingness uses all 891 labeled records; the fare histogram uses only the 712 training passengers. The right tail motivates a log transform, not automatic removal.

Age is missing for 177 passengers (19.9%), Cabin for 687 (77.1%), and Embarked for two (0.2%). Complete-case deletion would leave only 183 passengers, discarding most of the available sample. There are no exact duplicate rows or duplicate PassengerId values. The audit finds no negative ages, fares, or relative counts and no unsupported class, sex, or port categories.

Fractional infant ages are plausible. Fifteen fares equal zero, and the maximum fare is 512.3292. Neither observation alone proves an error, so both zero and high fares are retained. Missing values and unusual observed values are treated separately.

Step 1 Cleaning with explicit decisions

The main cleaning problem is incomplete information rather than widespread invalid values. Median imputation preserves passengers with unknown ages and is less sensitive to extreme ages than a mean. The median is estimated only from the training partition: 28.5 years. Within cross-validation, each training fold estimates its own median. This avoids using held-out feature distributions to influence training [4].

Field or issue	Treatment	Reason
Missing Age	Training median plus AgeMissing flag	Retain rows while distinguishing imputed from observed ages.
Missing Embarked	Training mode, S	Only two missing labels; a frequent category is a simple baseline.
Sparse Cabin labels	HasCabin = recorded cabin indicator	Preserve recording information without fitting many rare cabin identifiers.
Fare skew and extremes	Keep values; use $\log(1 + \text{Fare})$	Reduce scale differences; preserve valid zero fares.
Spelling and whitespace	Strip spaces; standardize case	Prevent equivalent categories from splitting into separate labels.
Invalid ranges or categories	Convert to missing before imputation	Defensive rule; no such invalid entries were found in the supplied labeled data.

Observed before and after examples

PassengerId	Original values	After cleaning and engineering
693	Age missing; Fare 56.4958; Cabin missing	Age 28.5; AgeMissing 1; HasCabin 0; LogFare 4.052
482	Age missing; Fare 0; Cabin missing	Age 28.5; AgeMissing 1; HasCabin 0; LogFare 0
830	Age 62; Embarked missing; Cabin B28	Age 62 retained; Embarked S; HasCabin 1

These examples come from the training partition. The notebook also verifies that the final transformed training matrix contains only finite values. Fare imputation is included defensively because the separate competition test file contains a missing fare, even though the labeled training file does not. Missing numeric values use training medians, never test-set medians.

No rows are dropped after the split. Missing-cabin status may encode differences in recordkeeping and passenger class, so HasCabin is a predictive proxy rather than a direct measure of cabin quality. Median age is likewise a practical replacement, not a claim about any particular passenger's true age.

Step 2 Features and transformations

The transformations are chosen to match the model assumptions. Linear models can use continuous quantities after scaling. Bernoulli Naive Bayes expects binary indicators, so continuous quantities are discretized using bin edges fitted on training data. The same raw passengers and target are used for both approaches, while the representations differ for legitimate modeling reasons.

Predictor group	Linear Regression, Ridge, LASSO	Bernoulli Naive Bayes
Age and LogFare	Median imputation; standardization	Median imputation; three quantile bins; binary bin indicators
Pclass, Sex, Embarked	Mode imputation; one-hot encoding with one reference category dropped	Mode imputation; full one-hot encoding
FamilySize = SibSp + Parch + 1	Median imputation; standardization	Training quantile bins; repeated edges yield two bins
IsAlone, HasCabin, AgeMissing	Three 0/1 indicators	The same three 0/1 indicators

LogFare is $\log(1 + \text{Fare})$, which accepts zero and compresses the high-fare tail. Standardization subtracts the training mean and divides by the training standard deviation. It is especially relevant to Ridge and LASSO because their penalties act on coefficient magnitudes. Binary flags remain unscaled; consequently the regularization convention differs between standardized continuous features and binary contrasts.

FamilySize summarizes accompanying relatives. IsAlone allows a separate offset for passengers without relatives, while FamilySize still represents broader variation. HasCabin and AgeMissing capture whether information was recorded. The linear design has 11 features; the NB design has 19 binary features. Duplicate quantile edges reduce FamilySize to two effective bins. Exact fitted bin counts are printed in the notebook.

Feature ablation on training folds

Model	Basic CV accuracy	Engineered CV accuracy	Change
Bernoulli NB	77.39%	75.84%	-1.55 percentage points
Linear Regression	78.94%	80.48%	+1.54 percentage points

The basic design contains Age, LogFare, Pclass, Sex, and Embarked. The engineered design adds FamilySize and the three flags. The ablation is diagnostic: the planned engineered design remains the primary comparison. The NB decline is consistent with extra dependent information violating its independence approximation, although this experiment does not isolate a single cause. A future validation-based feature search could compare subsets without reusing the current holdout.

Step 3 Training and model selection

A stratified random split with seed 42 allocates 712 passengers to training and 179 to a holdout. Training contains 273 survivors; the holdout contains 69 survivors and 110 nonsurvivors. Five stratified folds within the training partition, also seeded at 42, are shared across every candidate. All imputers, encoders, bin edges, and scales are refitted within each fold. No holdout outcome is used for hyperparameter selection.

Accuracy is the primary selection metric. The final NB setting is chosen between the two smoothed candidates. The unsmoothed model is a diagnostic control. Ridge and LASSO settings are selected separately, then their best candidates are compared with ordinary Linear Regression and the selected NB fit. Ties retain the first candidate in the declared order; thus NB selects 0.01 when both smoothed settings tie.

Candidate	Mean CV accuracy	Fold SD	Mean CV AUC
NB $\alpha=0$	0.7584	0.0282	0.8273
NB $\alpha=0.01$	0.7584	0.0282	0.8273
NB $\alpha=1$	0.7584	0.0264	0.8275
Linear Regression	0.8048	0.0143	0.8627
Ridge $\alpha=0.1$	0.8048	0.0143	0.8628
Ridge $\alpha=1$	0.8076	0.0144	0.8630
Ridge $\alpha=10$	0.8048	0.0143	0.8640
Ridge $\alpha=100$	0.7865	0.0152	0.8619
LASSO $\alpha=0.0001$	0.8048	0.0143	0.8626
LASSO $\alpha=0.001$	0.8006	0.0235	0.8633
LASSO $\alpha=0.01$	0.8006	0.0248	0.8632
LASSO $\alpha=0.1$	0.7008	0.0243	0.8134

Ridge with $\alpha = 1$ has the highest mean CV accuracy, 0.8076, compared with 0.8048 for ordinary Linear Regression. The difference is small relative to fold variability and should not be presented as decisive superiority. Fold SD measures variation across validation folds, not a standard error or independent confidence interval.

Linear Regression minimizes squared error to the binary target [3]. Ridge adds an L2 coefficient penalty; LASSO adds an L1 penalty. All regression outputs are thresholded at 0.5, as the assignment requires. NB also uses 0.5 on its estimated survival probability. Scores are kept continuous for ROC AUC, and no decision threshold is tuned after seeing the holdout.

Step 3 Additive smoothing experiment

Bernoulli Naive Bayes estimates class frequencies and binary feature frequencies within each class. It then combines those likelihoods using Bayes' rule and a conditional-independence approximation. For each binary feature j and class c , the smoothed estimate is $(N(j,c) + \alpha) / (N(c) + 2\alpha)$. The denominator includes two pseudo-counts because the feature can be either absent or present.

The implementation uses BernoulliNB with binarize disabled because the preprocessor already produces binary values. Setting force_alpha to true makes $\alpha = 0$ an actual unsmoothed control [2]. $\alpha = 1$ gives Laplace smoothing; $\alpha = 0.01$ adds much weaker smoothing. Gaussian variance stabilization would not fulfill the assignment's additive count-smoothing requirement, so GaussianNB is not used.

NB setting	CV accuracy	Holdout accuracy	Holdout AUC
$\alpha = 0$ · unsmoothed	0.7584	0.7039	0.7839
$\alpha = 0.01$ · weak smoothing	0.7584	0.7039	0.7839
$\alpha = 1$ · Laplace smoothing	0.7584	0.7039	0.7847

Measured change relative to no smoothing

Setting	Changed holdout labels	Mean absolute probability change
$\alpha = 0.01$	0 of 179	0.0000106
$\alpha = 1$	0 of 179	0.0010546

Smoothing did not improve or worsen held-out accuracy, precision, recall, or F1. It slightly changed estimated probabilities, and $\alpha = 1$ slightly changed their ordering, raising AUC by approximately 0.0008. These are small changes; the experiment provides no basis for claiming a meaningful performance gain from smoothing on this representation.

The unsmoothed class-conditional feature probabilities range from 0.0706 to 0.8519 in the full training fit. None are exactly zero or one, and the notebook checks that every validation and held-out prediction is finite. Therefore, zero-frequency failure did not occur here. The coarse binning and common categorical levels provide observations of both binary states in both classes.

The general purpose of smoothing still matters: with rarer features or a smaller training sample, an unseen feature state can otherwise force a class likelihood to zero. Smoothing trades a little bias toward 0.5 for less extreme estimates. It is a stabilizer, not a guarantee of higher classification accuracy. This distinction explains why the required experiment can be informative even when predicted labels do not change.

Step 4 Held-out model comparison

The table below evaluates the predeclared models on the same 179 passengers. Positive-class metrics refer to survivors. Accuracy counts all correct predictions; precision measures reliability among predicted survivors; recall measures the fraction of actual survivors identified; F1 balances precision and recall. AUC measures ranking across thresholds rather than accuracy at the fixed 0.5 threshold.

Model	Accuracy	Precision	Recall	F1	AUC
Majority baseline	0.615	0.000	0.000	0.000	0.500
NB $\alpha=0$	0.704	0.600	0.696	0.644	0.784
NB $\alpha=0.01$	0.704	0.600	0.696	0.644	0.784
NB $\alpha=1$	0.704	0.600	0.696	0.644	0.785
Linear Regression	0.799	0.762	0.696	0.727	0.832
Ridge $\alpha=1$	0.799	0.762	0.696	0.727	0.832
LASSO $\alpha=0.0001$	0.799	0.762	0.696	0.727	0.832

Ridge correctly predicts 143 of 179 outcomes (79.9%), which is 18.4 percentage points above the majority baseline. Ordinary Linear Regression and LASSO also make 143 correct predictions, with the same classifications passenger by passenger. All three identify 48 of 69 survivors. NB also identifies 48 survivors but makes 32 false-positive predictions, compared with 15 for the regression models.

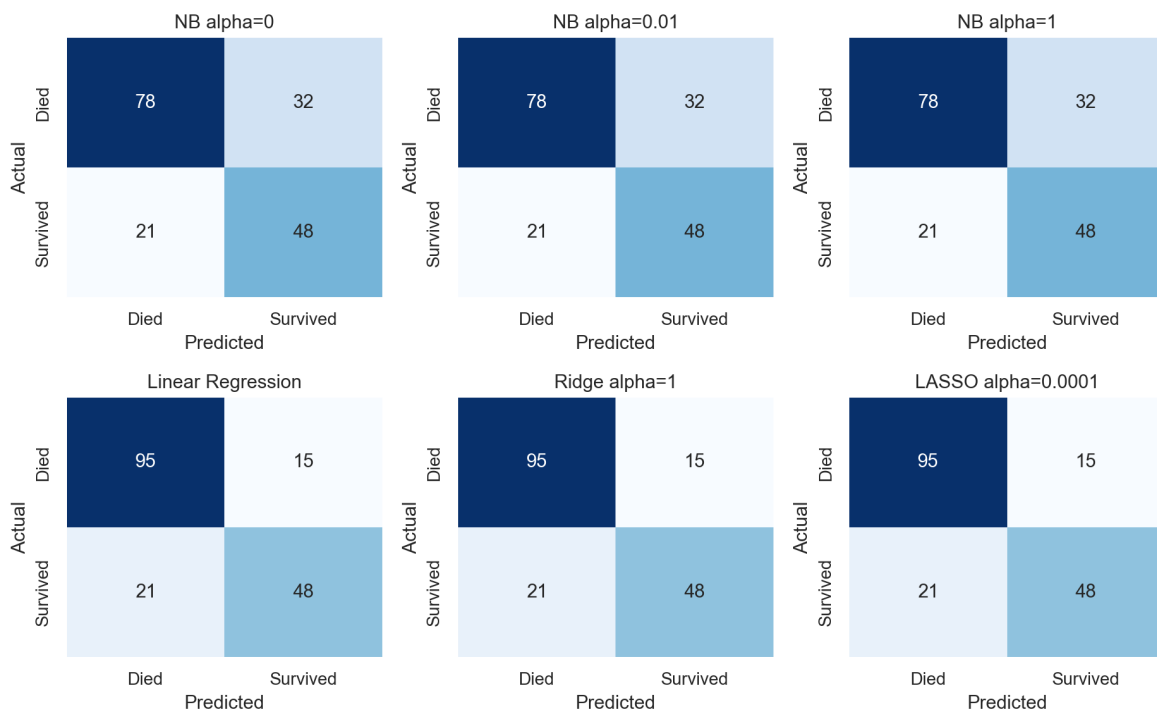


Figure 2. All six fitted-model confusion matrices use identical holdout records. Rows are actual outcomes and columns are predictions. NB makes 17 more false-positive errors; its false-negative count equals that of the regression models.

The baseline has zero survivor recall despite 61.5% accuracy. Class-specific metrics reveal this failure, which overall accuracy hides.

Step 4 Ranking and error interpretation

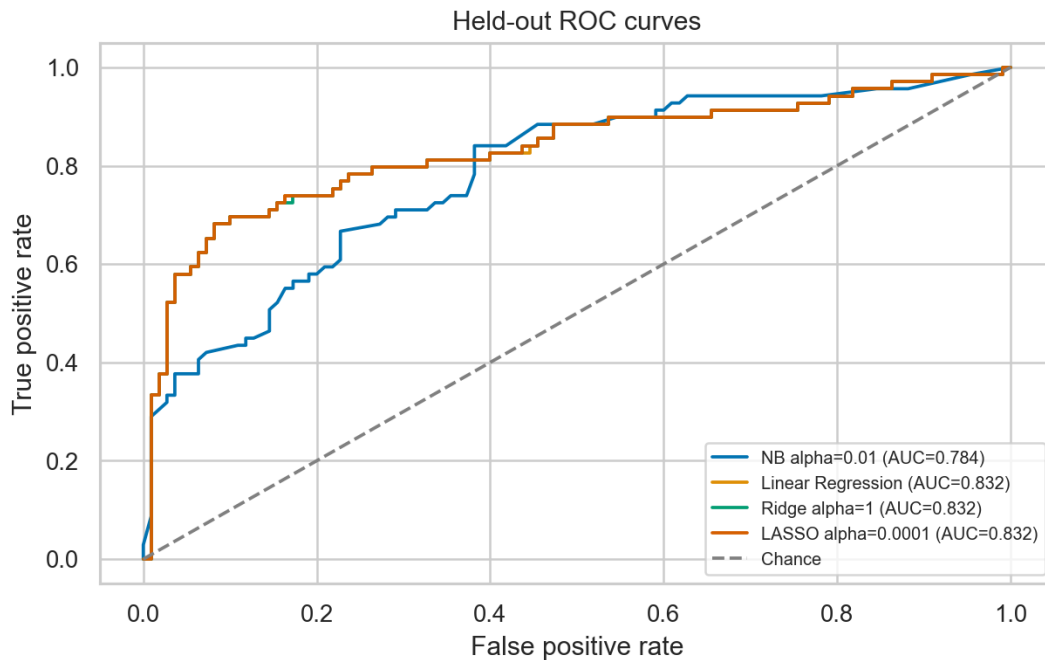


Figure 3. ROC curves use continuous survival scores. Ordinary Linear Regression and Ridge overlap closely. The diagonal is chance ranking. Regression scores need not be valid probabilities to compute ROC AUC.

The selected Ridge fit has AUC 0.8323, compared with 0.7839 for the selected NB fit. LASSO reaches 0.8324, a negligible difference in this holdout. An AUC around 0.83 indicates useful ranking in this sample; it does not imply that a score of 0.83 represents an 83% calibrated survival probability.

What the selected model gets wrong

Ridge produces 95 true negatives, 15 false positives, 21 false negatives, and 48 true positives. Its precision of 76.2% means that 48 of its 63 predicted survivors survived. Its recall of 69.6% means that 21 of 69 actual survivors were missed. If the application valued finding every survivor more than avoiding false positives, another threshold might be relevant, but the assignment fixes the threshold at 0.5.

Recorded sex	Holdout n	Accuracy	Survivor recall
Female	61	80.3%	100.0%
Male	118	79.7%	12.5%

Similar overall subgroup accuracies hide very different survivor recall. The holdout contains 45 female survivors and 24 male survivors; Ridge identifies all 45 female survivors but only three male survivors. This is consistent with a strong association between recorded sex and survival in the sample. It shows why aggregate accuracy alone can hide concentrated errors. These small historical subgroups do not support a broad fairness or causal conclusion.

The report uses the held-out errors descriptively. No feature, threshold, or hyperparameter is revised in response to this table, preserving its role as a final evaluation rather than an additional tuning set.

Regression behavior and regularization

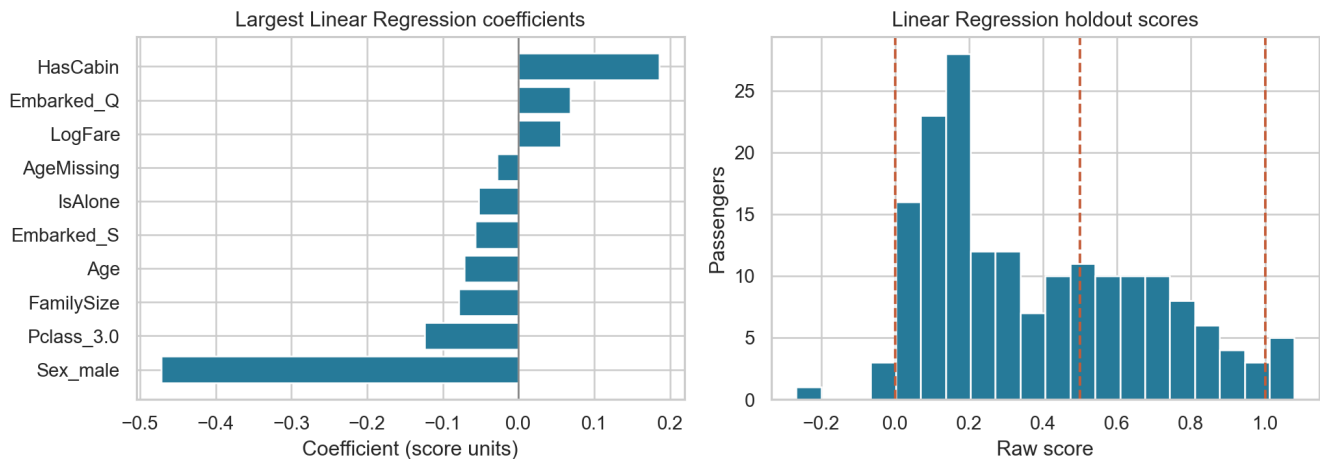


Figure 4. Linear Regression coefficients are associations conditional on the included predictors. Continuous predictors are standardized, while binary coefficients are category contrasts. The score histogram shows why regression outputs cannot be interpreted directly as probabilities.

Model	Score range	Outside [0,1]	Nonzero coefficients
Linear Regression	-0.267 to 1.079	10 / 179	11 / 11
Ridge $\alpha = 1$	-0.264 to 1.077	7 / 179	11 / 11
LASSO $\alpha = 0.0001$	-0.265 to 1.079	9 / 179	11 / 11

The required regression approach fits a straight score surface to a binary target, so scores below zero and above one are possible. Thresholding at 0.5 gives valid class labels but does not make the scores calibrated probabilities. Raw scores are retained for AUC because clipping can create ties and alter ranking. A model designed for binary outcomes, such as logistic regression, would be a reasonable later comparison but is not substituted for the required Linear Regression model.

Ridge shrinks the coefficient L2 norm from 0.5468 to 0.5435 at the selected setting. This modest change leaves every held-out class prediction unchanged. LASSO selects a very weak penalty and retains all 11 coefficients; sparse selection is possible with L1 regularization but did not occur at the selected setting. Stronger penalties reduced CV accuracy: Ridge at 100 reached 0.7865, and LASSO at 0.1 reached 0.7008.

Training accuracy is 81.2% for all three regression fits, versus 79.9% on the holdout. Their small observed gap does not prove absence of overfitting, but it gives no sign of a large train-test collapse in this split. NB training accuracy is 75.4% versus 70.4% on the holdout. Its lower performance may reflect information lost through discretization and dependent indicators; this comparison cannot isolate either mechanism.

Step 5 Discussion and limitations

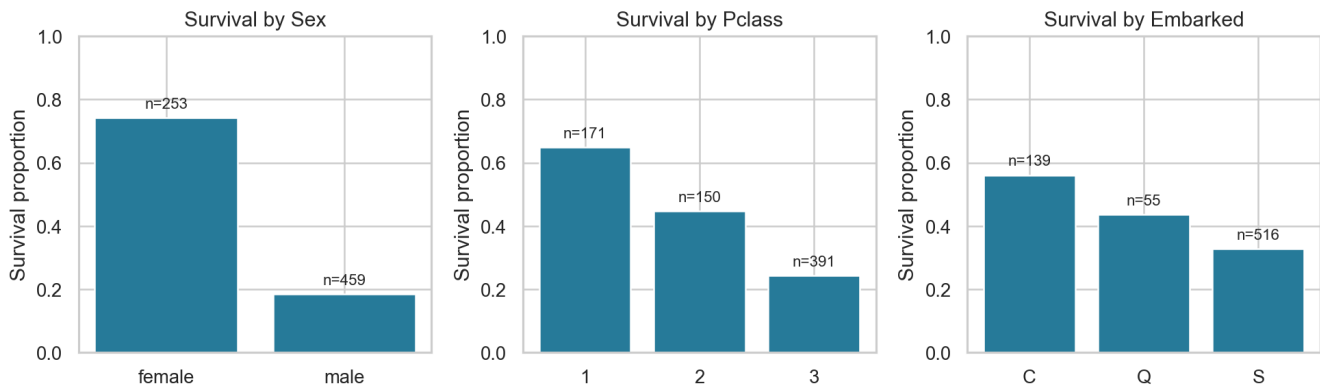


Figure 5. Descriptive survival rates are computed only on the training partition. Sample sizes are shown above bars; missing Embarked values are omitted from this descriptive grouping.

The training data show substantial survival differences by recorded sex and passenger class. These patterns explain why simple models can outperform a majority baseline, but they do not identify causal effects of changing any attribute. Fare, class, cabin recording, and group travel may overlap in what they represent. The strongest coefficients should therefore be interpreted as conditional associations within this design.

What this experiment supports

The selected Ridge model is a useful baseline for this split, and the required Linear Regression model performs equally well in held-out classification. The engineered features provide a modest training-CV gain for Linear Regression. NB is less effective with this feature representation; adding engineered predictors lowers its CV accuracy. Laplace smoothing regularizes its probabilities without improving its class predictions. Each conclusion follows the observed results rather than an assumption that more preprocessing or regularization must help.

What remains uncertain

The selected model's approximate 95% Wilson interval for held-out accuracy is 73.4% to 85.1%. This reflects the limited sample of 179 passengers and assumes independent outcomes. Relatives and passengers sharing tickets can appear on both sides of the random split, which may weaken that assumption and make generalization to unrelated groups harder than these results suggest.

Other limits include nonrandom missingness, a single historical sample, loss of detail through age imputation and NB binning, and potential differences between the labeled sample and other populations. Neither NB nor the regression fits were calibrated. Five-fold results used for selecting hyperparameters are not an unbiased external estimate of the selected model's performance; the reserved holdout provides the separate check.

Reproducibility and next experiment

The notebook records Python 3.10.7, NumPy 1.26.4, pandas 2.2.1, and scikit-learn 1.4.1.post1, together with seed 42. It displays metric tables and all five figures inline without separate image or CSV files. The optional final model is refitted on all 891 labeled rows and produces 418 competition predictions; no Kaggle score is claimed. A next study could use ticket or family groups for splitting and training-only validation to reduce correlated NB features, without reusing this holdout for tuning.

AI disclosure and references

AI tool usage disclosure

Tool used: OpenAI Codex, the ChatGPT coding assistant.

Contributions from AI: The assistant interpreted the assignment, inspected the supplied data, proposed the preprocessing and evaluation design, wrote the notebook, executed the models, checked the numerical outputs, generated visualizations, and drafted this report. It implemented the smoothing experiment, regularization comparison, feature ablation, subgroup diagnostics, and optional inline Kaggle predictions. The numerical findings come from executed analysis of the supplied CSV files.

Student contribution established in this session: The student selected Titanic, provided the local data and assignment instructions, and requested completion of the assignment. Independent coding, interpretation, and a full review by the student have not been verified. The student should update this disclosure after making any actual revisions and reviewing the work; AI-generated code or prose should not be represented as independently authored.

Required ownership review: Before submission, the student should be able to explain why missing ages are imputed, how train-only preprocessing prevents leakage, why Bernoulli NB needs binary features, what additive smoothing changes, why regression scores can exceed $[0,1]$, and why identical holdout labels do not establish that the models are equivalent in general. This follows the course handout's explicit instruction to understand each notebook block.

References

[1] Kaggle. Titanic — Machine Learning from Disaster. Dataset files and variable definitions.

<https://www.kaggle.com/competitions/titanic/data>

[2] scikit-learn documentation. BernoulliNB. Additive smoothing and force_alpha.

https://scikit-learn.org/stable/modules/generated/sklearn.naive_bayes.BernoulliNB.html

[3] scikit-learn documentation. LinearRegression. Ordinary least squares.

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

[4] scikit-learn documentation. Common pitfalls and recommended practices. Preprocessing and leakage.

https://scikit-learn.org/stable/common_pitfalls.html

Course source: Assignment 1: From Dirty Data to Predictive Models, supplied COMS4995W30 Applied Machine Learning handout. The handout specifies a runnable notebook, a 10–12 page report, a first-page Colab link and page map, both model families, smoothing comparisons, accuracy and confusion matrices, and an AI disclosure.

Submission readiness

The analysis and notebook are executed. The remaining external requirement is a real shareable Colab link on page 1. Upload the notebook, enable access, and replace the pending-link notice before submitting. Student review and any subsequent personal contributions should also be recorded accurately.